

VENTRICULAR AP

5 PHASES · IONS

- **Phase 0** — rapid depolarization, **Na⁺ influx** (fast voltage-gated Na channels).
- **Phase 1** — initial repolarization, Na channels inactivate + transient outward K⁺ current.
- **Phase 2** — plateau, **L-type Ca²⁺ influx** balances K⁺ efflux (cardiac-specific; gives the AP its long duration).
- **Phase 3** — rapid repolarization, Ca channels close + delayed-rectifier K⁺ efflux dominates.
- **Phase 4** — rest at -90 mV via inward-rectifier K⁺ (IK1) leak.
- **ERP = phases 0–3** — Na channels are inactivated; prevents tetany.
- **EC coupling:** phase-2 Ca²⁺ trigger → ryanodine receptor → *Ca-induced Ca release* from SR → troponin C → cross-bridges. Relaxation by **SERCA** (SR) + **NCX** (sarcolemma).

PACEMAKER AP

SA / AV NODES

- No stable resting potential — cells drift up from ~-60 mV until they spontaneously fire.
- **Phase 4** spontaneous depolarization — **HCN "funny" Na⁺ current** early, then T-type Ca²⁺, then L-type Ca²⁺ approaches threshold.
- **Phase 0** slow upstroke — **L-type Ca²⁺** ("slow-response" tissue).
- **Phase 3** repolarization — K⁺ efflux.
- Drug map: **β-blocker** ↓ HCN slope + L-Ca availability; **non-DHP CCB** blocks L-Ca directly; **adenosine** activates A₁ → ↑ K⁺ efflux → transient AV block; **ivabradine** selectively blocks HCN.
- Digoxin: blocks Na/K ATPase → ↑ intracellular Na → less NCX Ca extrusion → ↑ cytosolic Ca → ↑ contractility; vagomimetic AV slowing.

CONDUCTION · VELOCITY · ESCAPE RATES

PARK-AT-VENTURA-AV

- **Speed:** Purkinje (~4 m/s) > atrial myocardium > ventricular myocardium > AV node (~0.05 m/s).
- AV nodal delay (~100 ms) preserves the atrial kick and filters supraventricular tachycardias.
- Rate control in AFib all works by **prolonging AV nodal refractoriness** (β-blocker, non-DHP CCB, digoxin).

Pacemaker	Intrinsic rate	Escape QRS width
SA node	60–100 / min	Narrow (normal)
AV junction / His	40–60 / min	Narrow escape (block above bifurcation)
Purkinje / ventricular	20–40 / min	Wide escape (block below bifurcation; unstable)

ECG BASICS + HEART BLOCKS

4-STEP READ

- **Rate** 300/big-boxes between R waves (regular); count R's in 6 s × 10 (irregular).
- **Rhythm** sinus = P before every QRS, upright in II.
- **Axis:** I+ / aVF+ → normal · I+ / aVF- → LAD · I- / aVF+ → RAD · I- / aVF- → extreme.
- **Intervals:** PR 0.12–0.20 s · QRS < 0.12 s · QTc < 0.44 s (M) / 0.46 s (F).
- **Blocks:** 1° = long PR · Mobitz I = progressive PR → drop (AV node, benign) · Mobitz II = constant PR → random drop (infranodal, pacer) · 3° = AV dissociation (pacer).

ARRHYTHMIA GALLERY

RECOGNIZE ON A STRIP

- **AFib:** irregularly irregular, no P waves; foci in **pulmonary veins**.
- **Atrial flutter:** sawtooth flutter waves, atrial ~300/min, often 2:1 ventricular response; circuit around tricuspid annulus.
- **PSVT (AVNRT):** regular narrow tachy 150–220; no visible P (AVNRT) or retrograde P (AVRT). Vagal → adenosine.
- **PVC:** wide QRS premature beat + opposite-direction T + compensatory pause; post-PVC beat has ↑ SV from extra filling.
- **Monomorphic VT:** wide regular tachy 150–250; ICD for sustained in structural heart disease.
- **Complete AV block:** P-QRS dissociation; narrow escape (junctional, better) vs wide (ventricular, worse).

Arrhythmia		Anatomic ablation target	
AFib		Pulmonary vein isolation	
Atrial flutter (typical)		Cavotricuspid isthmus	
AVNRT		Slow pathway of AV node	
WPW		Accessory pathway (bundle of Kent)	
Idiopathic VT		RVOT (right ventricular outflow tract)	

Class	Target	AP duration	Examples
IA	Na block + some K block	Prolongs	Quinidine, procainamide, disopyramide
IB	Na block (ischemic tissue)	Shortens	Lidocaine (IV), mexiletine
IC	Strong Na block	No change	Flecainide, propafenone — <i>avoid in structural heart disease</i>
II	β -blockade	—	Metoprolol, esmolol, carvedilol
III	K ⁺ block	Prolongs (+ QT)	Amiodarone, sotalol, dofetilide, ibutilide
IV	L-type Ca block	—	Verapamil, diltiazem
Other	A ₁ / Mg / vagal	—	Adenosine (AVNRT), magnesium (torsades), digoxin (rate control)

CORONARY AUTOREGULATION + L-ARGININE → CGMP

- Heart extracts >70% O₂ at rest → **demand met by ↑ flow**, not extraction.
- Coronary flow autoregulated over MAP 60–140 mmHg; below 60 → ischemia; above 140 → passive flow.
- Mediators: **adenosine** (ATP breakdown product, A₂A vasodilation) + **nitric oxide**.
- NO biology**: eNOS uses **L-arginine** + O₂ + NADPH → **NO** + citrulline → smooth-muscle guanylate cyclase → **↑ cGMP** → vasodilation. PDE5 inhibitors (sildenafil) potentiate — *never combine with nitrates*.
- LV perfused mostly in **diastole** (intramural systolic pressure compresses vessels). **Subendocardium** fails first — deepest in the squeeze, last to perfuse.

BAROREFLEX · CHEMORECEPTORS · SYNCOPE

CAROTID IX · AORTIC X

- Carotid sinus** baroreceptor → afferent **CN IX** → solitary nucleus.
- Aortic arch** baroreceptor → afferent **CN X**. (Mnemonic: *carotid IX, aortic X*.)
- Carotid body + aortic body chemoreceptors (also IX / X) respond to ↓ PaO₂, ↑ PaCO₂, ↓ pH.
- ↑ BP** → ↑ stretch → ↑ firing → ↑ vagal + ↓ sympathetic → ↓ HR / contractility / vasoconstriction.
- ↓ BP / shock** → reverse: ↑ HR + ↑ SVR + ↓ vagal tone (Block-2 hemorrhagic shock pattern).
- Vasovagal syncope** = brainstem misfires both arms: *cardioinhibitory* (↑ vagal → ↓ HR) + *vasodepressor* (↓ sympathetic → vasodilation). Trigger (pain, heat, prolonged standing), prodrome (warmth, pallor, nausea), brief LOC, rapid recovery.
- Carotid sinus hypersensitivity**: tight collar / shaving in an older adult → exaggerated reflex bradycardia + hypotension.

SHOCK — FOUR TYPES · HEMODYNAMIC PROFILE

FA 320–321

Type	CVP	PCWP	CI	SVR	SvO ₂	Skin	Examples
Hypovolemic	↓	↓	↓	↑	↓	cold	Hemorrhage, GI losses, burns
Cardiogenic	↑	↑	↓	↑	↓	cold	STEMI with LV failure, acute decompensated HF
Obstructive	↑	↓ (PE) / ↑ (tamponade)	↓	↑	↓	cold	PE, cardiac tamponade, tension pneumothorax
Distributive	↓	↓	↑*	↓	↑*	warm	Septic, anaphylactic, neurogenic

- * Distributive: CO and SvO₂ high in septic / anaphylactic; neurogenic may be lower from lost sympathetic tone.
- Tamponade PCWP can read paradoxically ↑ (transmitted external pericardial pressure); LV is actually under-filled.
- Acute pediatric diarrhea → hypovolemia + lactic acidosis + Kussmaul respirations (compensatory respiratory alkalosis).

- ↓ SV / CO triggers four arms: **sympathetic** (↑ HR, contractility, vasoconstriction), **RAAS** (AngII + aldosterone + remodeling), **ADH** (water retention → dilutional hyponatremia — poor prognostic marker), **ANP / BNP** counter-regulation.
- **ANP** from atrial myocytes (stretch); **BNP** from ventricular myocytes. Both activate guanylate cyclase → ↑ cGMP → natriuresis + vasodilation + ↓ aldosterone + ↓ renin. Cleared by **neprilysin**.
- **Sacubitril** inhibits neprilysin → ↑ endogenous ANP/BNP. Paired with **valsartan** (ARB) to block RAAS independently → "ARNi" (sacubitril/valsartan).

Pillar	Drug examples	Mechanism / effect
β-blocker	Carvedilol, metoprolol succinate, bisoprolol	Blocks sympathetic arm; reverses remodeling; mortality benefit
ACE-i / ARB / ARNi	Lisinopril, valsartan, sacubitril/valsartan	Blocks RAAS; ARNi additionally ↑ ANP/BNP
MRA	Spironolactone, eplerenone	Blocks aldosterone; ↓ K/Mg loss; ↓ fibrosis
SGLT2 inhibitor	Dapagliflozin, empagliflozin	Reduces HF hospitalization regardless of diabetes

- Loop diuretic (furosemide) — symptomatic congestion relief only; no mortality benefit.
- Acute decompensated HF with HTN → nitroprusside (balanced; SV preserved).
- BNP: useful to rule out acute HF (< 100); falsely low in obesity; unreliable on ARNi → use NT-proBNP.

TOP 15 TRAPS

1. Ventricular phase 0 = Na⁺; pacemaker phase 0 = Ca²⁺.
2. HCN "funny" current drives pacemaker phase 4.
3. Conduction speed: Park-At-Ventura, AV (Purkinje > atria > ventricles > AV node).
4. Carotid → CN IX; aortic → CN X (baro and chemo).
5. AFib triggers in pulmonary veins; flutter circuit at cavotricuspid isthmus.
6. Vagal maneuvers + adenosine break AVNRT via prolonged AV refractoriness.
7. Coronary sinus = lowest O₂ in body; LV perfused in diastole; subendocardium fails first.
8. NO substrate = L-arginine; mediator = cGMP.
9. Digoxin: Na/K ATPase block → ↑ cytosolic Ca via NCX → ↑ contractility; vagomimetic AV slowing.
10. Cardiogenic shock SVR is *up* (compensatory), not down.
11. Distributive shock = ↑ CO + ↓ SVR (except neurogenic).
12. Tamponade PCWP reads ↑ on cath but LV is under-filled.
13. Sacubitril blocks neprilysin → ↑ ANP/BNP; valsartan blocks AT₁.
14. Complete AV block: narrow QRS escape (His, better); wide escape (ventricular, worse).
15. Class IC drugs (flecainide, propafenone) —